Number Systems

Digital System Fundamentals

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Learning Outcome

- Understand binary, octal, and hexadecimal numbers.
- Convert numbers amongst decimal, binary, octal and hexadecimal.
- Calculate binary arithmetic.

Contents

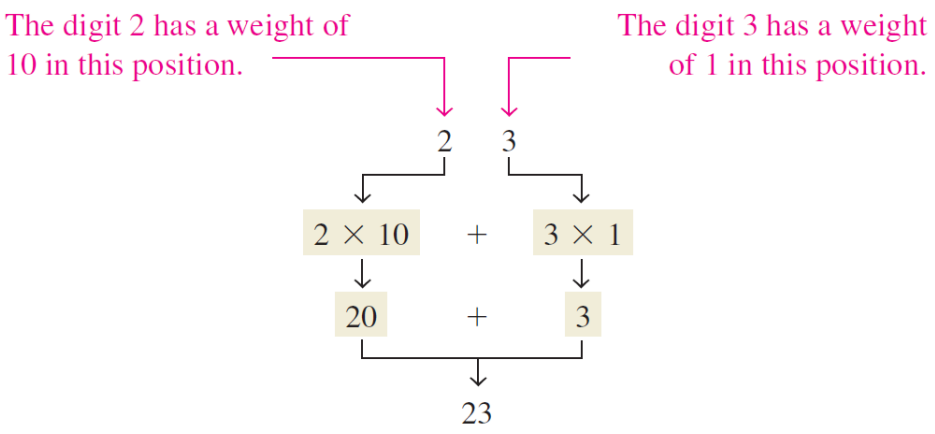
1. Decimal Numbers
2. Binary Numbers
3. Binary Arithmetic
4. Complements of Binary Numbers
5. Hexadecimal Numbers
6. Octal Numbers
7. Software

1. Decimal Numbers

Decimal Number System (Base-ten System)

- Familiar
- Ten symbols / digits
  - 0 1 2 3 4 5 6 7 8 9
- Combine more digits
  - Different position,

    Different magnitude / weight
- Power of Ten



$$\ldots\ 10^5\ 10^4\ 10^3\ 10^2\ 10^1\ 10^0$$

$$10^2\ 10^1\ 10^0.10^{-1}\ 10^{-2}\ 10^{-3}\ \ldots$$

Decimal point

Decimal Numbers - Example

EXAMPLE 2–1

Express the decimal number 47 as a sum of the values of each digit.

Solution

The digit 4 has a weight of 10, which is 10^1, as indicated by its position. The digit 7 has a weight of 1, which is 10^0, as indicated by its position.

$$\begin{aligned} 47 &= (4 \times 10^1) + (7 \times 10^0) \\ &= (4 \times 10) + (7 \times 1) = \mathbf{40 + 7} \end{aligned}$$

Related Problem*

Determine the value of each digit in 939.

*Answers are at the end of the chapter.

Decimal Numbers - Example

EXAMPLE 2–2

Express the decimal number 568.23 as a sum of the values of each digit.

Solution

The whole number digit 5 has a weight of 100, which is 10^2, the digit 6 has a weight of 10, which is 10^1, the digit 8 has a weight of 1, which is 10^0, the fractional digit 2 has a weight of 0.1, which is 10^{-1}, and the fractional digit 3 has a weight of 0.01, which is 10^{-2}.

$$\begin{aligned} 568.23 &= (5 \times 10^2) + (6 \times 10^1) + (8 \times 10^0) + (2 \times 10^{-1}) + (3 \times 10^{-2}) \\ &= (5 \times 100) + (6 \times 10) + (8 \times 1) + (2 \times 0.1) + (3 \times 0.01) \\ &= \mathbf{500} + \mathbf{60} + \mathbf{8} + \mathbf{0.2} + \mathbf{0.03} \end{aligned}$$

Related Problem

Determine the value of each digit in 67.924.

Decimal Numbers - Example

SECTION 2–1 CHECKUP

Answers are at the end of the chapter.

1. What weight does the digit 7 have in each of the following numbers?

   (a) 1370 **(b)** 6725 **(c)** 7051 **(d)** 58.72

2. Express each of the following decimal numbers as a sum of the products obtained by multiplying each digit by its appropriate weight:

   (a) 51 **(b)** 137 **(c)** 1492 **(d)** 106.58

2. Binary Numbers

Binary Numbers (Base-two System)

- 2 symbols / digits
  - 0 1
  - Binary Digit - Bit
- Combine more bits
  - Different position,

    Different magnitude / weight
- Power of Two

$$2^{n-1} \ldots 2^3\, 2^2\, 2^1\, 2^0 \,.\, 2^{-1}\, 2^{-2} \ldots 2^{-n}$$

Binary point

Binary Number - Counting

- For binary *n* bits
- Largest decimal number
  - 2^n-1
- LSB : Least Significant Bit
- MSB : Most Significant Bit

TABLE 2–1

Decimal Number	Binary Number			
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

Binary Numbers - Weight

TABLE 2–2

Binary weights.

Positive Powers of Two (Whole Numbers)									Negative Powers of Two (Fractional Number)					
2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}	2^{-4}	2^{-5}	2^{-6}
256	128	64	32	16	8	4	2	1	1/2 0.5	1/4 0.25	1/8 0.125	1/16 0.625	1/32 0.03125	1/64 0.015625

Binary to Decimal Conversion

EXAMPLE 2–3

Convert the binary whole number 1101101 to decimal.

Solution

Determine the weight of each bit that is a 1, and then find the sum of the weights to get the decimal number.

$$\begin{aligned} \text{Weight:} \quad & 2^6\ 2^5\ 2^4\ 2^3\ 2^2\ 2^1\ 2^0 \\ \text{Binary number:} \quad & 1\ \ 1\ \ 0\ \ 1\ \ 1\ \ 0\ \ 1 \\ 1101101 &= 2^6 + 2^5 + 2^3 + 2^2 + 2^0 \\ &= 64 + 32 + 8 + 4 + 1 = \mathbf{109} \end{aligned}$$

Related Problem

Convert the binary number 10010001 to decimal.

Binary to Decimal Conversion

EXAMPLE 2–4

Convert the fractional binary number 0.1011 to decimal.

Solution

Determine the weight of each bit that is a 1, and then sum the weights to get the decimal fraction.

Weight:	2^{-1}	2^{-2}	2^{-3}	2^{-4}
Binary number:	0 . 1	0	1	1

$$0.1011 = 2^{-1} + 2^{-3} + 2^{-4}$$
$$= 0.5 + 0.125 + 0.0625 = \mathbf{0.6875}$$

Related Problem

Convert the binary number 10.111 to decimal.

SECTION 2–2 CHECKUP

1. What is the largest decimal number that can be represented in binary with eight bits?
2. Determine the weight of the 1 in the binary number 10000.
3. Convert the binary number 10111101.011 to decimal.

Decimal to Binary Conversion

Sum-of-Weights Method

To get the binary number for a given decimal number, find the binary weights that add up to the decimal number.

$$9 = 8 + 1 \quad \text{or} \quad 9 = 2^3 + 2^0$$

2^3	2^2	2^1	2^0	
1	0	0	1	Binary number for decimal 9

Convert the following decimal numbers to binary:

(a) 12 **(b)** 25

(c) 58 **(d)** 82

TABLE 2–2

Binary weights.

Positive Powers of Two (Whole Numbers)									Negative Powers of Two (Fractional Number)					
2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}	2^{-4}	2^{-5}	2^{-6}
256	128	64	32	16	8	4	2	1	1/2 0.5	1/4 0.25	1/8 0.125	1/16 0.625	1/32 0.03125	1/64 0.015625

Decimal to Binary Conversion

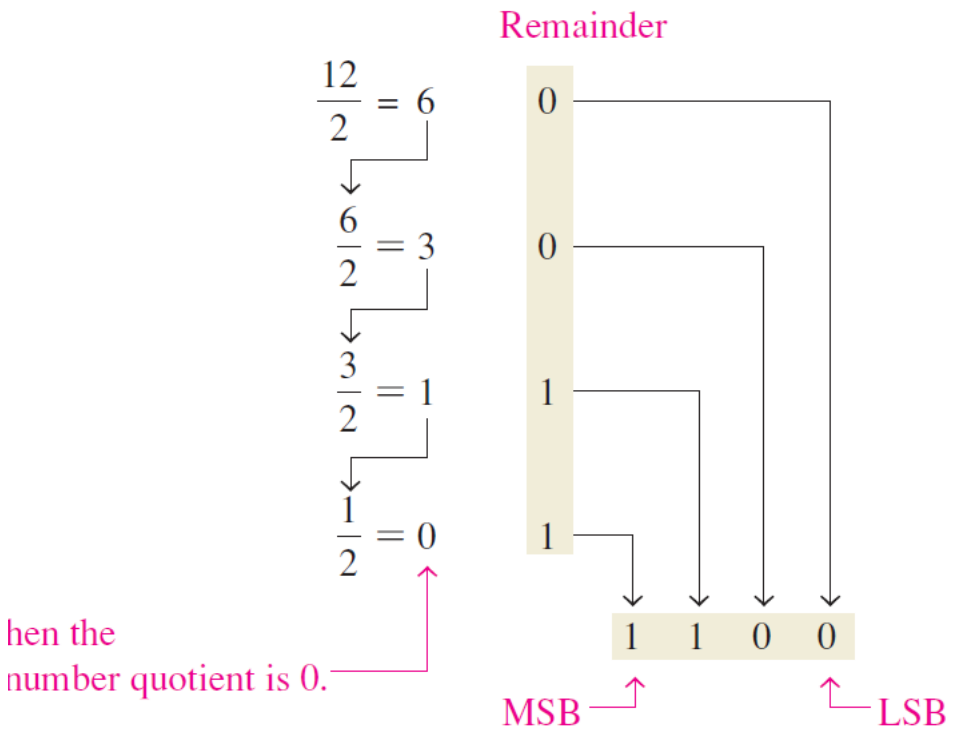
Repeated Division-by-2 Method

To get the binary number for a given decimal number, divide the decimal number by 2 until the quotient is 0. Remainders form the binary number.

Convert the following decimal numbers to binary:

(a) 12 **(b)** 25

(c) 58 **(d)** 82



Converting Decimal Fractions to Binary

Sum-of-Weights

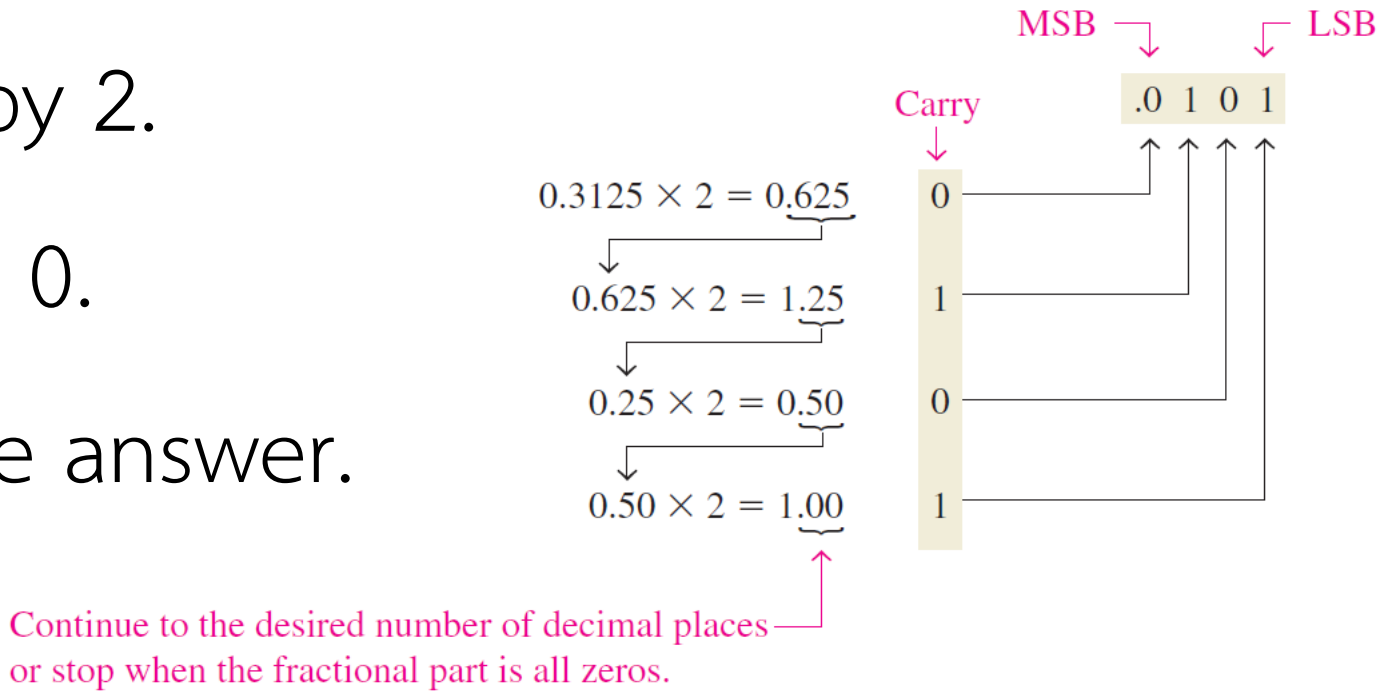
The sum-of-weights method can be applied to fractional decimal numbers, as shown in the following example:

$$0.625 = 0.5 + 0.125 = 2^{-1} + 2^{-3} = 0.101$$

There is a 1 in the 2^{-1} position, a 0 in the 2^{-2} position, and a 1 in the 2^{-3} position.

Converting Decimal Fractions to Binary

- Multiplication by 2.
- Until fraction is 0.
- Carries form the answer.



Decimal to Binary Conversion

SECTION 2–3 CHECKUP

1. Convert each decimal number to binary by using the sum-of-weights method:

   (a) 23 (b) 57 (c) 45.5

2. Convert each decimal number to binary by using the repeated division-by-2 method (repeated multiplication-by-2 for fractions):

   (a) 14 (b) 21 (c) 0.375

TABLE 2–2

Binary weights.

Positive Powers of Two (Whole Numbers)									Negative Powers of Two (Fractional Number)					
2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}	2^{-4}	2^{-5}	2^{-6}
256	128	64	32	16	8	4	2	1	1/2 0.5	1/4 0.25	1/8 0.125	1/16 0.625	1/32 0.03125	1/64 0.015625

Practice 2.1 | Number System Conversion

1. Convert decimal to binary

    1.1) 27 (Sum-of-Weights Method)

    1.2) 48 (Repeated Division-by-2 Method)

    1.3) 12.425 (Only 8 bit for fraction)

2. Convert binary to decimal to binary

    2.1) 1101

    2.2) 101.111

    2.3) 10011101

3. Binary Arithmetic

Addition, Subtraction, Multiplication and Division

Binary Addition

- Results are *sum* and *carry.*
- Four basic rules

$0 + 0 = 0$	Sum of 0 with a carry of 0
$0 + 1 = 1$	Sum of 1 with a carry of 0
$1 + 0 = 1$	Sum of 1 with a carry of 0
$1 + 1 = 10$	Sum of 0 with a carry of 1

Carry bits

$1 + 0 + 0 = 01$	Sum of 1 with a carry of 0
$1 + 1 + 0 = 10$	Sum of 0 with a carry of 1
$1 + 0 + 1 = 10$	Sum of 0 with a carry of 1
$1 + 1 + 1 = 11$	Sum of 1 with a carry of 1

Binary Addition

EXAMPLE 2–7

Add the following binary numbers:

(a) 11 + 11 **(b)** 100 + 10

(c) 111 + 11 **(d)** 110 + 100

Binary Addition

EXAMPLE 2–7

Add the following binary numbers:

(a) 11 + 11 **(b)** 100 + 10

(c) 111 + 11 **(d)** 110 + 100

Solution

The equivalent decimal addition is also shown for reference.

(a)
$$\begin{array}{r} 11 \\ +\ 11 \\ \hline \mathbf{110} \end{array} \qquad \begin{array}{r} 3 \\ +\ 3 \\ \hline 6 \end{array}$$

(b)
$$\begin{array}{r} 100 \\ +\ 10 \\ \hline \mathbf{110} \end{array} \qquad \begin{array}{r} 4 \\ +\ 2 \\ \hline 6 \end{array}$$

(c)
$$\begin{array}{r} 111 \\ +\ 11 \\ \hline \mathbf{1010} \end{array} \qquad \begin{array}{r} 7 \\ +\ 3 \\ \hline 10 \end{array}$$

(d)
$$\begin{array}{r} 110 \\ +\ 100 \\ \hline \mathbf{1010} \end{array} \qquad \begin{array}{r} 6 \\ +\ 4 \\ \hline 10 \end{array}$$

Related Problem

Add 1111 and 1100.

Binary Subtraction

- Four basic rules

$0 - 0 = 0$	
$1 - 1 = 0$	
$1 - 0 = 1$	
$10 - 1 = 1$	$0 - 1$ with a borrow of 1

EXAMPLE 2–8

Perform the following binary subtractions:

(a) $11 - 01$ **(b)** $11 - 10$

Solution

(a)

$$\begin{array}{r} 11 \\ -01 \\ \hline \mathbf{10} \end{array} \qquad \begin{array}{r} 3 \\ -1 \\ \hline 2 \end{array}$$

(b)

$$\begin{array}{r} 11 \\ -10 \\ \hline \mathbf{01} \end{array} \qquad \begin{array}{r} 3 \\ -2 \\ \hline 1 \end{array}$$

No borrows were required in this example. The binary number 01 is the same as 1.

Related Problem

Subtract 100 from 111.

Binary Subtraction

EXAMPLE 2–9

Subtract 011 from 101.

Solution

$$\begin{array}{r} 101 \\ -011 \\ \hline \mathbf{010} \end{array} \qquad \begin{array}{r} 5 \\ -3 \\ \hline 2 \end{array}$$

Let's examine exactly what was done to subtract the two binary numbers since a borrow is required. Begin with the right column.

Left column:
When a 1 is borrowed,
a 0 is left, so $0 - 0 = 0$.

Middle column:
Borrow 1 from next column
to the left, making a 10 in
this column, then $10 - 1 = 1$.

$$\begin{array}{r} {}^{0}\cancel{1}^{1}01 \\ -0\,11 \\ \hline 0\,10 \end{array}$$

Right column:
$1 - 1 = 0$

Related Problem

Subtract 101 from 110.

Binary Multiplication

- Four basic rules

$$0 \times 0 = 0$$
$$0 \times 1 = 0$$
$$1 \times 0 = 0$$
$$1 \times 1 = 1$$

EXAMPLE 2–10

Perform the following binary multiplications:

(a) 11×11 **(b)** 101×111

Solution

(a)

$$\begin{array}{r} 11 \\ \times\ 11 \\ \hline 11 \\ +11 \\ \hline \mathbf{1001} \end{array} \qquad \begin{array}{r} 3 \\ \times\ 3 \\ \hline 9 \end{array}$$

Partial products

(b)

$$\begin{array}{r} 111 \\ \times\ 101 \\ \hline 111 \\ 000 \\ +111 \\ \hline \mathbf{100011} \end{array} \qquad \begin{array}{r} 7 \\ \times\ 5 \\ \hline 35 \end{array}$$

Partial products

Related Problem

Multiply 1101×1010.

Binary Division

EXAMPLE 2–11

Perform the following binary divisions:

(a) 110 ÷ 11 **(b)** 110 ÷ 10

Solution

(a)

```
      10        2
11)110      3)6
   11         6
   ---        ---
   000        0
```

(b)

```
      11        3
10)110      2)6
   10         6
   ---        ---
    10        0
    10
    ---
    00
```

Related Problem

Divide 1100 by 100.

Practice 2.2 | Binary Arithmetic

SECTION 2–4 CHECKUP

1. Perform the following binary additions:

   (a) 1101 + 1010 (b) 10111 + 01101

2. Perform the following binary subtractions:

   (a) 1101 − 0100 (b) 1001 − 0111

3. Perform the indicated binary operations:

   (a) 110 × 111 (b) 1100 ÷ 011

4. Complements of Binary Numbers

Complements of Binary Numbers

- Represent of negative number / signed number
  - Sign-Magnitude
  - 1’s Complement
  - 2’s Complement

Sign-Magnitude

The Sign Bit

The left-most bit in a signed binary number is the **sign bit**, which tells you whether the number is positive or negative.

A 0 sign bit indicates a positive number, and a 1 sign bit indicates a negative number.

Sign-Magnitude Form

When a signed binary number is represented in sign-magnitude, the left-most bit is the sign bit and the remaining bits are the magnitude bits. The magnitude bits are in true (uncomplemented) binary for both positive and negative numbers. For example, the decimal number +25 is expressed as an 8-bit signed binary number using the sign-magnitude form as

$$0\underbrace{0011001}_{\text{Magnitude bits}}$$

Sign bit

The decimal number −25 is expressed as

$$10011001$$

Notice that the only difference between +25 and −25 is the sign bit because the magnitude bits are in true binary for both positive and negative numbers.

In the sign-magnitude form, a negative number has the same magnitude bits as the corresponding positive number but the sign bit is a 1 rather than a zero.

TABLE 2–1

Decimal Number	Binary Number			
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

1's Complement

Finding the 1's Complement

The 1's **complement** of a binary number is found by changing all 1s to 0s and all 0s to 1s, as illustrated below:

1 0 1 1 0 0 1 0	Binary number
↓↓↓↓↓↓↓↓	
0 1 0 0 1 1 0 1	1's complement

The simplest way to obtain the 1's complement of a binary number with a digital circuit is to use parallel inverters (NOT circuits), as shown in Figure 2–2 for an 8-bit binary number.

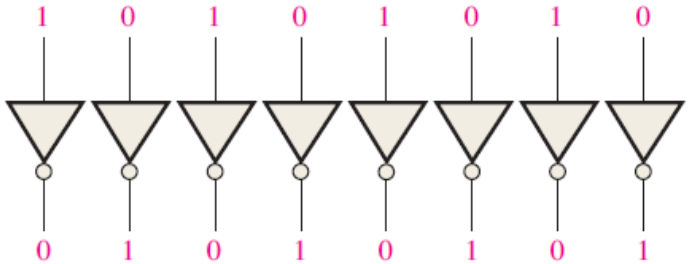


FIGURE 2–2 Example of inverters used to obtain the 1's complement of a binary number.

TABLE 2–1

Decimal Number	Binary Number			
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

2's Complement

Finding the 2's Complement

The 2's complement of a binary number is found by adding 1 to the LSB of the 1's complement.

$$\text{2's complement} = (\text{1's complement}) + 1$$

EXAMPLE 2–12

Find the 2's complement of 10110010.

Solution

10110010	Binary number
01001101	1's complement
+ 1	Add 1
01001110	2's complement

Related Problem

Determine the 2's complement of 11001011.

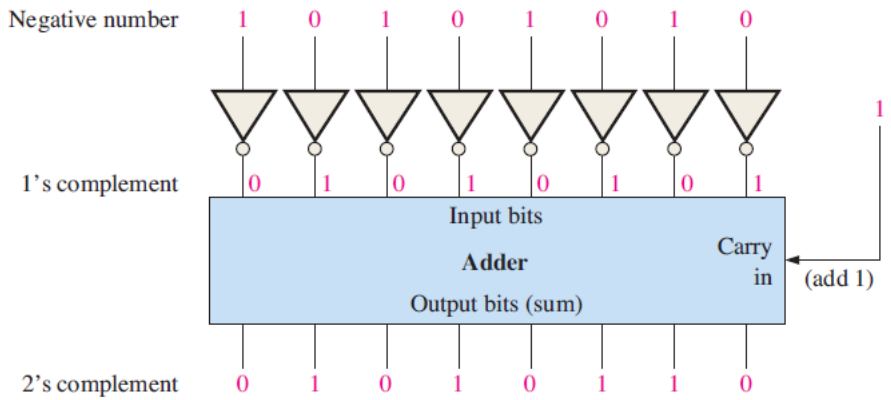


FIGURE 2–3 Example of obtaining the 2's complement of a negative binary number.

SECTION 2–5 CHECKUP

1. Determine the 1's complement of each binary number:
   (a) 00011010 (b) 11110111 (c) 10001101
2. Determine the 2's complement of each binary number:
   (a) 00010110 (b) 11111100 (c) 10010001

2's Complement

TABLE 2–1

Decimal Number	Binary Number			
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

3 Forms of Signed Numbers

EXAMPLE 2–14

Express the decimal number −39 as an 8-bit number in the sign-magnitude, 1's complement, and 2's complement forms.

Solution

First, write the 8-bit number for +39.

00100111

In the *sign-magnitude form,* −39 is produced by changing the sign bit to a 1 and leaving the magnitude bits as they are. The number is

10100111

In the *1's complement form,* −39 is produced by taking the 1's complement of +39 (00100111).

11011000

In the *2's complement form,* −39 is produced by taking the 2's complement of +39 (00100111) as follows:

$$\begin{array}{rl} 11011000 & \text{1's complement} \\ +\quad\quad 1 & \\ \hline \mathbf{11011001} & \text{2's complement} \end{array}$$

Related Problem

Express +19 and −19 as 8-bit numbers in sign-magnitude, 1's complement, and 2's complement.

Determine Decimal Value of Signed Numbers

- Sign-Magnitude
  - 10010101, 01110111
- 1's Complement
  - 00010111, 11101000, 11101011
- 2's Complement
  - 01010110, 10101010, 11010111
  - 11, 111, 1111, 11111, 11111111

Range of 2's Complement

- Binary n bits
- Binary range : 0 to 2^n-1
- 2's Complement range : $-(2^{n-1})$ to $+(2^{n-1} - 1)$
- Binary 4 bits
- Binary range : 0 – 15
- 2's Complement range : (-8) – (+7)

Subtraction with 2's Complement

To subtract two signed numbers, take the 2's complement of the subtrahend and add.

Discard any final carry bit.

EXAMPLE 2–20

Perform each of the following subtractions of the signed numbers:

(a) 00001000 − 00000011 (b) 00001100 − 11110111

(c) 11100111 − 00010011 (d) 10001000 − 11100010

Solution

Like in other examples, the equivalent decimal subtractions are given for reference.

(a) In this case, $8 - 3 = 8 + (-3) = 5$.

	00001000	Minuend (+8)
	+ 11111101	2's complement of subtrahend (−3)
Discard carry →	**1 00000101**	Difference (+5)

(b) In this case, $12 - (-9) = 12 + 9 = 21$.

00001100	Minuend (+12)
+ 00001001	2's complement of subtrahend (+9)
00010101	Difference (+21)

(c) In this case, $-25 - (+19) = -25 + (-19) = -44$.

	11100111	Minuend (−25)
	+ 11101101	2's complement of subtrahend (−19)
Discard carry	**1 11010100**	Difference (−44)

(d) In this case, $-120 - (-30) = -120 + 30 = -90$.

10001000	Minuend (−120)
+ 00011110	2's complement of subtrahend (+30)
10100110	Difference (−90)

Related Problem

Subtract 01000111 from 01011000.

Practice 2.3 | Signed Numbers

1. Convert -12_{10} (-01100_2) to the following forms:

    1.1 Sign magnitude

    1.2 1's complement

    1.3 2's complement

2. Perform each arithmetic in 2's complement form and give decimal equivalences

    2.1 00010110 + 00110011

    2.2 01110000 + 10101111

    2.3 00110011 – 00010000

    2.4 01100101 - 11101000

5. Hexadecimal Numbers

Hexadecimal Numbers

- 16 Symbols
- 1 HEX digit represents 4-bit binary.
- Power of Sixteen

TABLE 2–3

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Counting in Hexadecimal

. . ., E, F, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 1A, 1B, 1C, 1D, 1E, 1F, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 2A, 2B, 2C, 2D, 2E, 2F, 30, 31, . . .

Binary-to-Hexadecimal Conversion

EXAMPLE 2–24

Convert the following binary numbers to hexadecimal:

(a) 1100101001010111 **(b)** 111111000101101001

Solution

(a) $\underbrace{1100}_{\downarrow\,C}\underbrace{1010}_{\downarrow\,A}\underbrace{0101}_{\downarrow\,5}\underbrace{0111}_{\downarrow\,7} = \mathbf{CA57}_{16}$

(b) $\underbrace{0011}_{\downarrow\,3}\underbrace{1111}_{\downarrow\,F}\underbrace{0001}_{\downarrow\,1}\underbrace{0110}_{\downarrow\,6}\underbrace{1001}_{\downarrow\,9} = \mathbf{3F169}_{16}$

Two zeros have been added in part (b) to complete a 4-bit group at the left.

Related Problem

Convert the binary number 1001111011110011100 to hexadecimal.

Hexadecimal-to-Binary Conversion

EXAMPLE 2–25

Determine the binary numbers for the following hexadecimal numbers:

(a) $10A4_{16}$ **(b)** $CF8E_{16}$ **(c)** 9742_{16}

Solution

(a) 1 0 A 4 → **1000010100100**

(b) C F 8 E → **1100111110001110**

(c) 9 7 4 2 → **1001011101000010**

In part (a), the MSB is understood to have three zeros preceding it, thus forming a 4-bit group.

Related Problem

Convert the hexadecimal number 6BD3 to binary.

Hexadecimal is a ***convenient*** way to represent binary numbers.

Hexadecimal-to-Decimal Conversion

EXAMPLE 2–26

Convert the following hexadecimal numbers to decimal:

(a) $1C_{16}$ **(b)** $A85_{16}$

Solution

Remember, convert the hexadecimal number to binary first, then to decimal.

(a)

$$\overbrace{\overset{1}{\downarrow}}^{0001}\overbrace{\overset{C}{\downarrow}}^{1100} = 2^4 + 2^3 + 2^2 = 16 + 8 + 4 = \mathbf{28}_{10}$$

(b)

$$\overbrace{\overset{A}{\downarrow}}^{1010}\overbrace{\overset{8}{\downarrow}}^{1000}\overbrace{\overset{5}{\downarrow}}^{0101} = 2^{11} + 2^9 + 2^7 + 2^2 + 2^0 = 2048 + 512 + 128 + 4 + 1 = \mathbf{2693}_{10}$$

Related Problem

Convert the hexadecimal number 6BD to decimal.

Hexadecimal-to-Decimal Conversion

EXAMPLE 2–27

Convert the following hexadecimal numbers to decimal:

(a) $E5_{16}$ **(b)** $B2F8_{16}$

Solution

Recall from Table 2–3 that letters A through F represent decimal numbers 10 through 15, respectively.

(a) $E5_{16} = (E \times 16) + (5 \times 1) = (14 \times 16) + (5 \times 1) = 224 + 5 = \mathbf{229}_{10}$

(b)
$$\begin{aligned} B2F8_{16} &= (B \times 4096) + (2 \times 256) + (F \times 16) + (8 \times 1) \\ &= (11 \times 4096) + (2 \times 256) + (15 \times 16) + (8 \times 1) \\ &= 45{,}056 + 512 + 240 + 8 = \mathbf{45{,}816}_{10} \end{aligned}$$

Related Problem

Convert $60A_{16}$ to decimal.

Decimal-to-Hexadecimal Conversion

EXAMPLE 2–28

Convert the decimal number 650 to hexadecimal by repeated division by 16.

Solution

Hexadecimal remainder

$$\frac{650}{16} = 40.625 \rightarrow 0.625 \times 16 = 10 = \text{A}$$

$$\frac{40}{16} = 2.5 \rightarrow 0.5 \times 16 = 8 = 8$$

$$\frac{2}{16} = 0.125 \rightarrow 0.125 \times 16 = 2 = 2$$

Stop when whole number quotient is zero.

2 8 A Hexadecimal number

MSD LSD

Related Problem

Convert decimal 2591 to hexadecimal.

6. Octal Numbers

Octal Numbers

- 8 Symbols
  - 0 1 2 3 4 5 6 7
- Power of Eight
- One octal digit represents three-bit binary.
- Counting

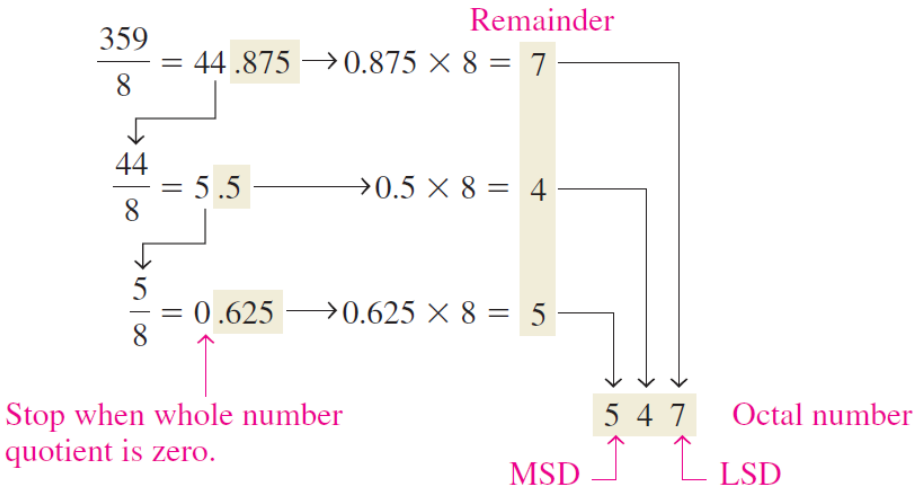
$$10, 11, 12, 13, 14, 15, 16, 17, 20, 21, \ldots$$

Conversion

Octal-to-Decimal Conversion

$$
\begin{aligned}
&\text{Weight:} && 8^3\ 8^2\ 8^1\ 8^0 \\
&\text{Octal number:} && 2\ \ 3\ \ 7\ \ 4 \\
2374_8 &= (2 \times 8^3) + (3 \times 8^2) + (7 \times 8^1) + (4 \times 8^0) \\
&= (2 \times 512) + (3 \times 64) + (7 \times 8) + (4 \times 1) \\
&= 1024 + 192 + 56 + 4 = 1276_{10}
\end{aligned}
$$

Decimal-to-Octal Conversion



Octal-to-Binary Conversion

TABLE 2–4

Octal/binary conversion.

Octal Digit	0	1	2	3	4	5	6	7
Binary	000	001	010	011	100	101	110	111

EXAMPLE 2–31

Convert each of the following octal numbers to binary:

(a) 13_8 **(b)** 25_8 **(c)** 140_8 **(d)** 7526_8

Solution

(a) 1 3 → **001011**

(b) 2 5 → **010101**

(c) 1 4 0 → **001100000**

(d) 7 5 2 6 → **111101010110**

Related Problem

Convert each of the binary numbers to decimal and verify that each value agrees with the decimal value of the corresponding octal number.

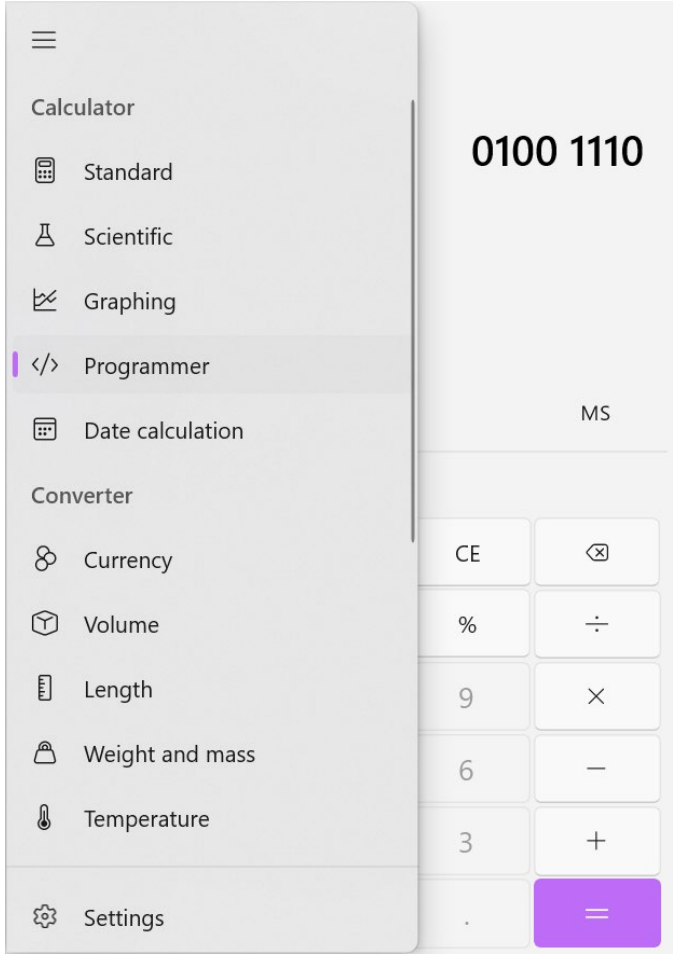
Practice 2.4 | HEX&OCT

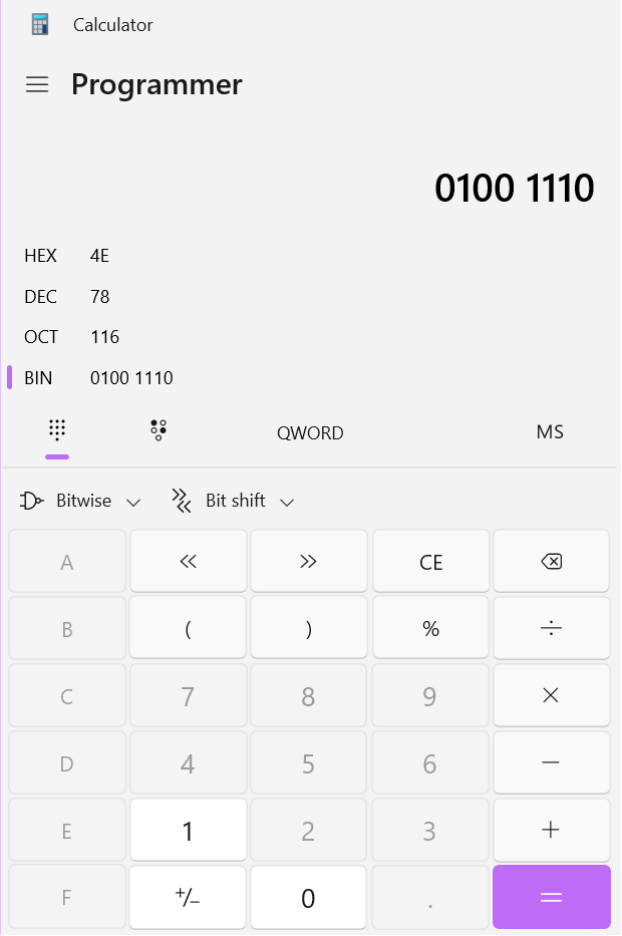
1. Convert 1110001110101111_2 to hexadecimal and octal

2. Convert 46_{16} and $ABCD_{16}$ to binary

7. Conversion & Calculation Software

Calculator - Windows





Number Conversion - Website

RapidTables

Home › Conversion › Number conversion › Decimal to binary

Decimal to Binary converter

From: Decimal

To: Binary

Enter decimal number

123 10

= Convert | × Reset | ⇅ Swap

Binary number

1111011 2

Binary signed 2's complement

0000000001111011 2

Hex number

7B 16

Decimal to binary calculation steps

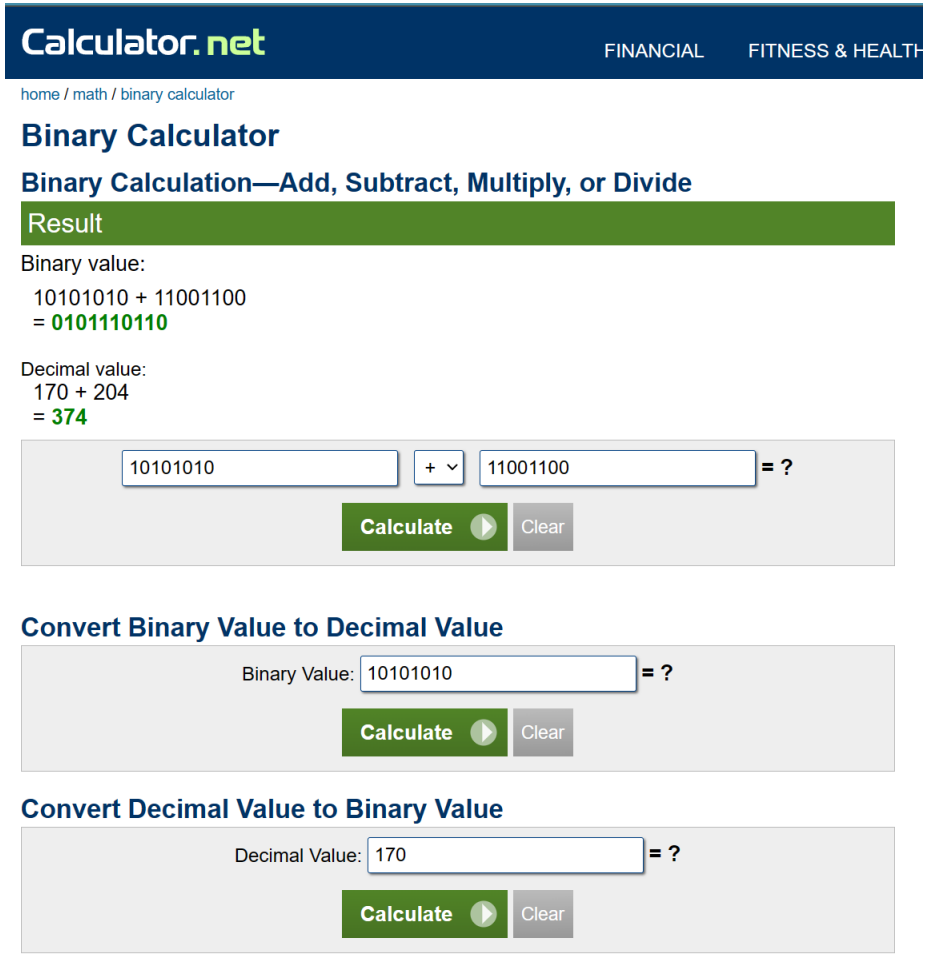
Divide by the base 2 to get the digits from the remainders:

Division by 2	Quotient	Remainder (Digit)	Bit #
(123)/2	61	1	0
(61)/2	30	1	1
(30)/2	15	0	2
(15)/2	7	1	3
(7)/2	3	1	4
(3)/2	1	1	5
(1)/2	0	1	6

$= (1111011)_2$

Decimal to Binary Converter (rapidtables.com)

Binary Calculator - Website



Binary Calculator

THANK YOU